EXP1	Implementation of fuzzy control inference system

AIM: Understand the concept of fuzzy control inference system using python programming language.

Algorithm:
Step 1: Define Fuzzy Sets input and output variables.
Step 2: Create Fuzzy Rules
Step 3: Perform Fuzzy Inference
Step 4: Defuzzify the output fuzzy sets to obtain a crisp output value.
Step 5: Use the defuzzified output as the control action.
Step 6: Implement Control Action.
Step 7: Repeat the above steps in a loop as needed for real-time control.
End of the fuzzy control algorithm.

First, you'll need to install the scikit-fuzzy library if you haven't already. You can install it using the following command:

```
pip install scikit-fuzzy
```

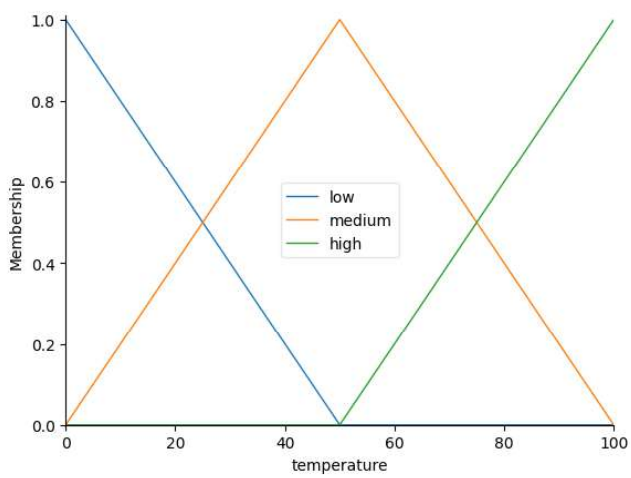
Now, let's implement the fuzzy inference system:

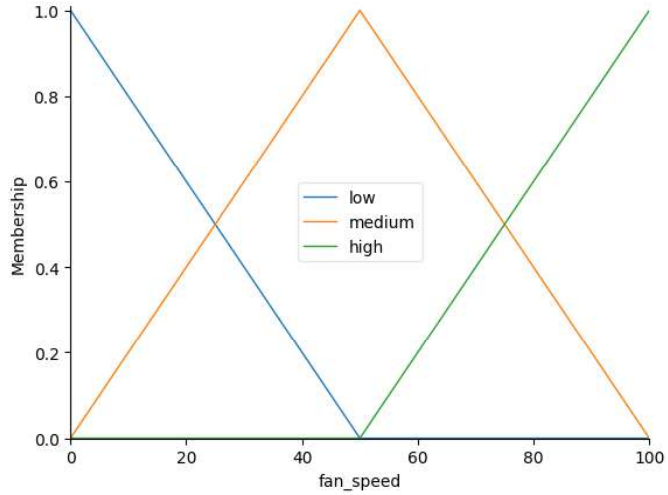
PROGRAM:

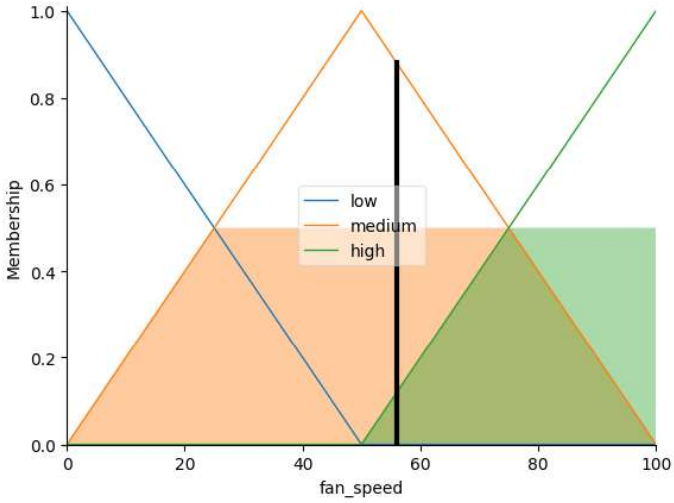
```
import numpy as np
import skfuzzy as fuzz
from skfuzzy import control as ctrl

# Create Antecedent/Consequent objects for temperature and fan speed
temperature = ctrl.Antecedent(np.arange(0, 101, 1), 'temperature')
fan_speed = ctrl.Consequent(np.arange(0, 101, 1), 'fan_speed')

# Define membership functions for temperature
temperature['low'] = fuzz.trimf(temperature.universe, [0, 0, 50])
temperature['medium'] = fuzz.trimf(temperature.universe, [0, 50, 100])
temperature['high'] = fuzz.trimf(temperature.universe, [50, 100, 100])

# Define membership functions for fan speed
fan_speed['low'] = fuzz.trimf(fan_speed.universe, [0, 0, 50])
fan_speed['medium'] = fuzz.trimf(fan_speed.universe, [0, 50, 100])
fan_speed['high'] = fuzz.trimf(fan_speed.universe, [50, 100, 100])

# Define fuzzy rules
rule1 = ctrl.Rule(temperature['low'], fan_speed['low'])
```

```
rule2 = ctrl.Rule(temperature['medium'], fan_speed['medium'])
rule3 = ctrl.Rule(temperature['high'], fan_speed['high'])

# Create control system and add rules
fan_ctrl = ctrl.ControlSystem([rule1, rule2, rule3])
fan_speed_ctrl = ctrl.ControlSystemSimulation(fan_ctrl)

# Input the temperature value
temperature_value = 75

# Pass the input to the control system
fan_speed_ctrl.input['temperature'] = temperature_value

# Compute the result
fan_speed_ctrl.compute()


# Print the output
print("Fan Speed:", fan_speed_ctrl.output['fan_speed'])
# Plot membership functions and output
temperature.view()
fan_speed.view()
fan_speed.view(sim=fan_speed_ctrl)
```

Output:







Result: Thus the above program for fuzzy control interface system executed successfully with desired output.